

Luke Ramsdell

designer + engineer + maker

Work

Smithsonian National Air and Space Museum

Exhibits Intern

June - August 2025

UNC Charlotte

Teaching Assistant,

Instrumentation Lab

August 2025 - Present

NC State Design Lab

Weekend Workshop Instructor

Fall & Spring 2024

NC State Design Lab

Design Camp Counselor & TA

June - August 2024

Husqvarna Group

Industrial Design Intern

May - December 2023

Skills

Mastered Softwares

Blender • SolidWorks • Illustrator

Photoshop • InDesign

Proficient Softwares

Enscape • KeyShot

Autodesk Fusion & CAMplete

Matlab • Python

Premier Pro • Microsoft Office

Practical Skills

FDM 3D Printing • Laser Cutting

Programming for CNC Milling

Basic Wood & Metalworking

Foam & Resin Modelmaking

Vacuum Forming • Sewing

Circuits and Instrumentation

Sketching • Prototyping

Visualization • Presentation

Education

NC State University

Bachelor's of Industrial Design

2025 - GPA 4.0

University of NC at Charlotte

Master's of Science in

Mechanical Enigneering

In Progress - GPA 4.0

This Portfolio

■ **Next Summer**

What I bring to the table in 2026

■ **Beyond the Museum**

My skills in other projects

■ **Last Summer**

A look back: 10 weeks at NASM

Next Summer



What I bring to the table in 2026

Project 1:
3D Scan **Auto-Processing**

An experiment in improving 3D workflows
through tailor-made tools.

The Challenge:

build bespoke 3D tools to solve unique problems and improve niche workflows

↓ Blender's Python API offers a versatile sandbox for building and testing custom solutions like this one

```
vert = workVerts.new(fullVertCos)
ring.append(vert)

rings.append(ring)

for thisRing in range(nPlanes - 1):
    for thisVert in range(nCleanVerts):
        #grab the four verts to be used in making a face
        v0 = rings[thisRing][thisVert]
        v1 = rings[thisRing][(thisVert + 1) % nCleanVerts] #modulo makes it wrap around to 0 at the end
        v2 = rings[thisRing + 1][(thisVert + 1) % nCleanVerts]
        v3 = rings[thisRing + 1][thisVert]

        #create the faces
        workBM.faces.new((v0, v1, v2, v3))

##push the new bmesh to a new object##
cleanMe = bpy.data.meshes.new("light mesh")
workBM.to_mesh(cleanMe)
cleanMe.update()
workBM.free()
cleanObj = bpy.data.objects.new("light mesh", cleanMe)
bpy.context.scene.collection.objects.link(cleanObj)

class cleanScanOp(Operator): #create an operator; important: name must be unique from functions, property groups, etc.
    '''build a coarse mesh with elliptical cross-sections approximating selected vertices'''

    bl_idname = "collection.cleanscanop" #attach the operator to a portion of the blender data and give it an idname (must be all lowercase)
    bl_label = "Build Light Mesh"

    bl_options = {"REGISTER", "UNDO"} #Make it undoable

    def execute(self, context): #execute method (needed for operator to do things)
        props = context.scene.scanCleanProps

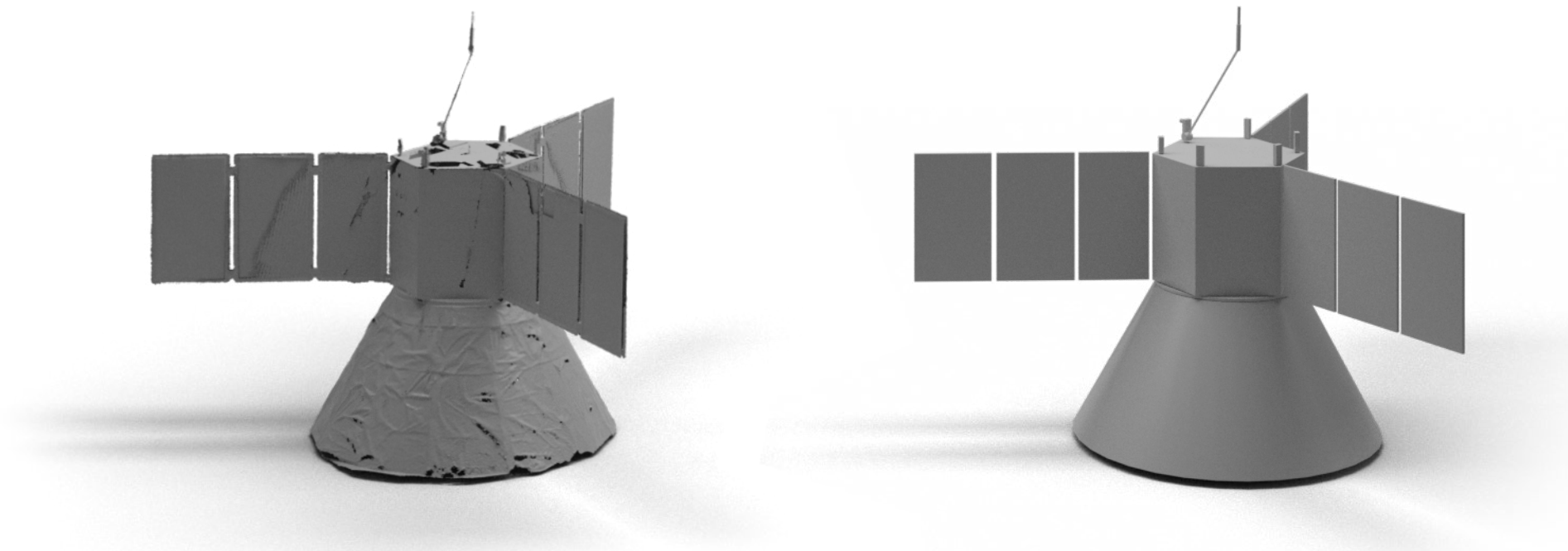
        selectedVerts = getSelectedVerts()
```


A NASM Problem: 3D-Scanned Artifacts

Raw 3D scans are **too data-heavy** to be used in gallery drawings directly.

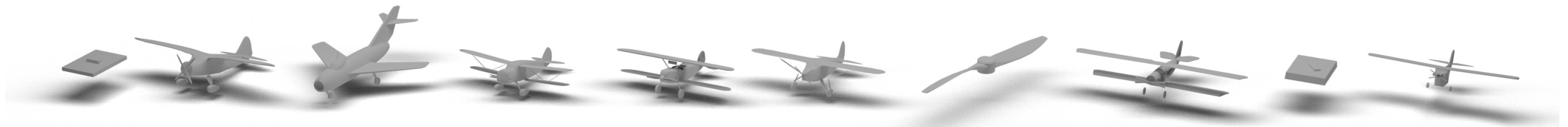
Lighter (“low-poly”) models approximating their overall shape and size must be made.

Currently, this is a **manual process**.

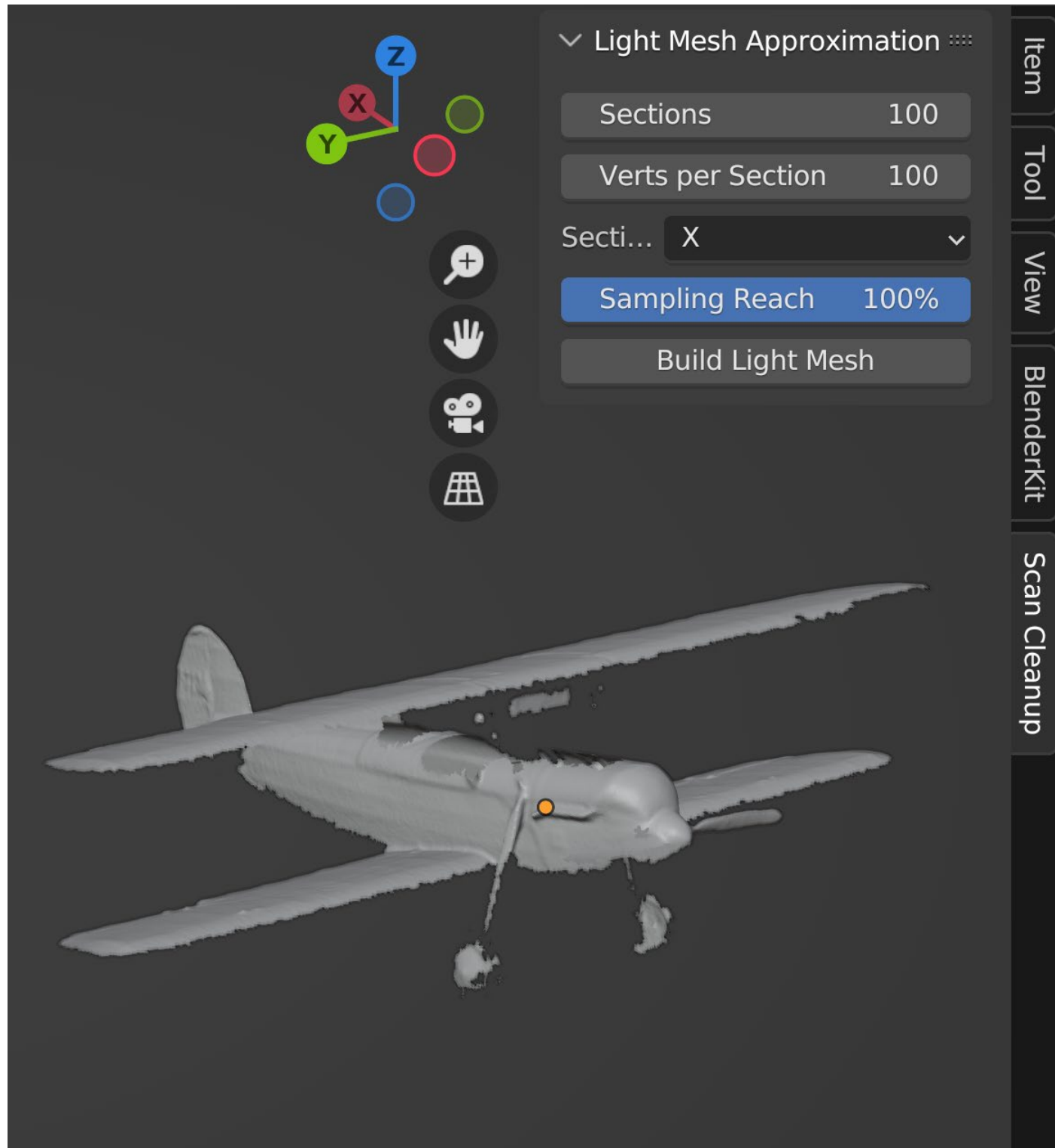


Heavy 3D Scan → **Manual Process** → Light Model

↓ Light 3D models of artifacts that I built manually last summer. Even with an efficient Blender workflow, this is time consuming for NASM’s large collection.

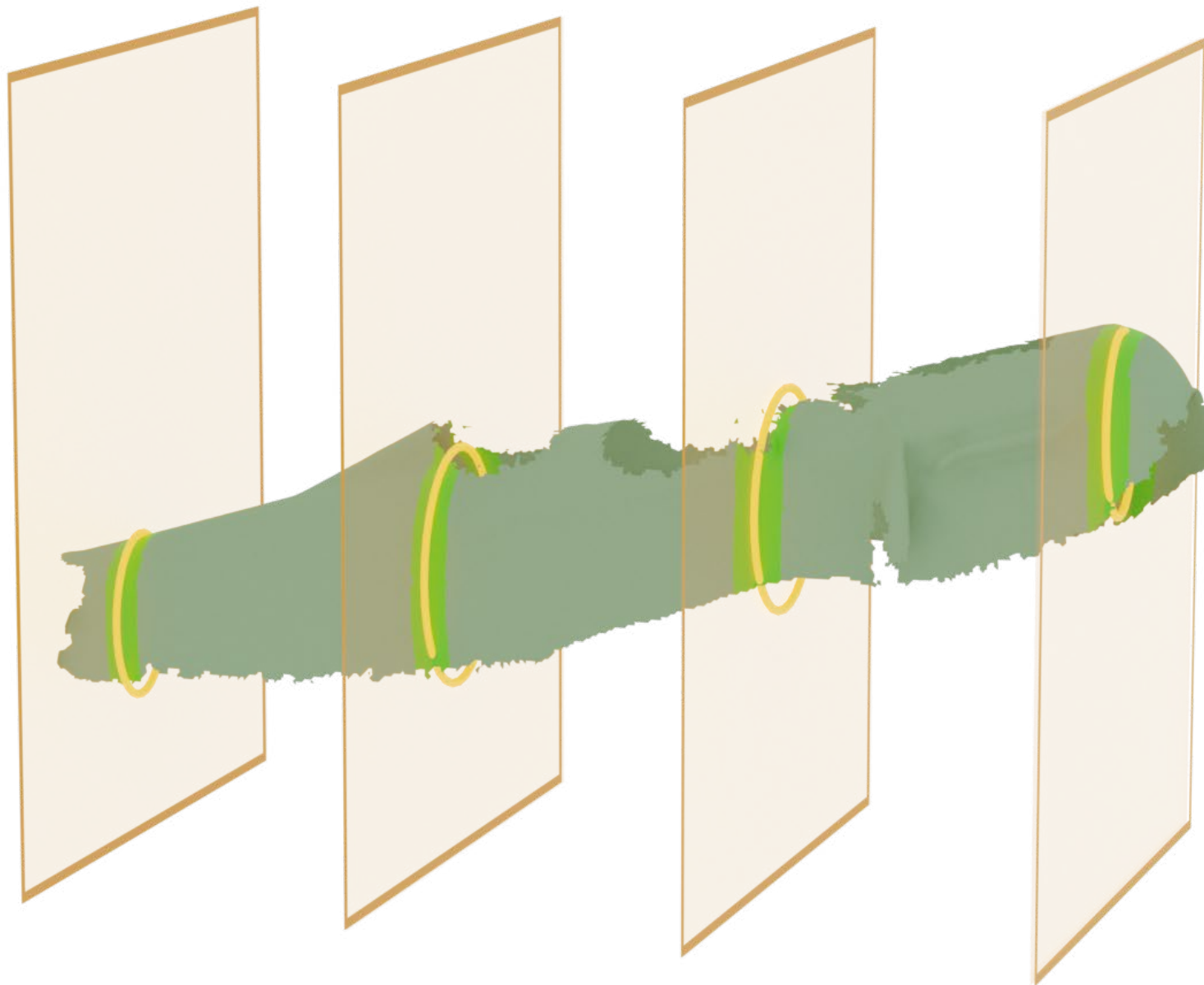


The Solution: **A Custom 3D Tool**



Using Python code and Blender's mesh modeling capabilities, I created a **fully custom tool** to **automate** part of the **3D scan re-modeling process**.

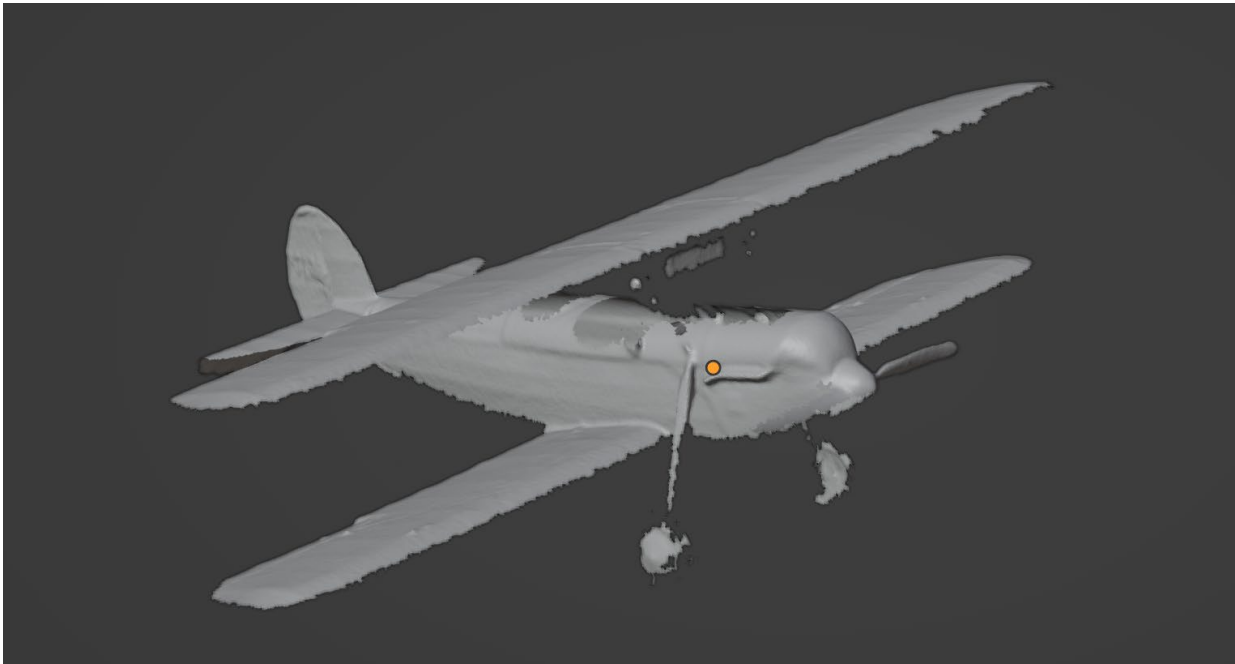
How it Works



The tool samples the **3D scan** around a series of **planes**, then mathematically fits **elliptical cross-sections** to the **sampled geometry**. These cross-sections are connected to form the light model.

This approach effectively approximates the **streamlined forms** common in the NASM collection, and naturally **fills holes** which are common in raw 3D scans.

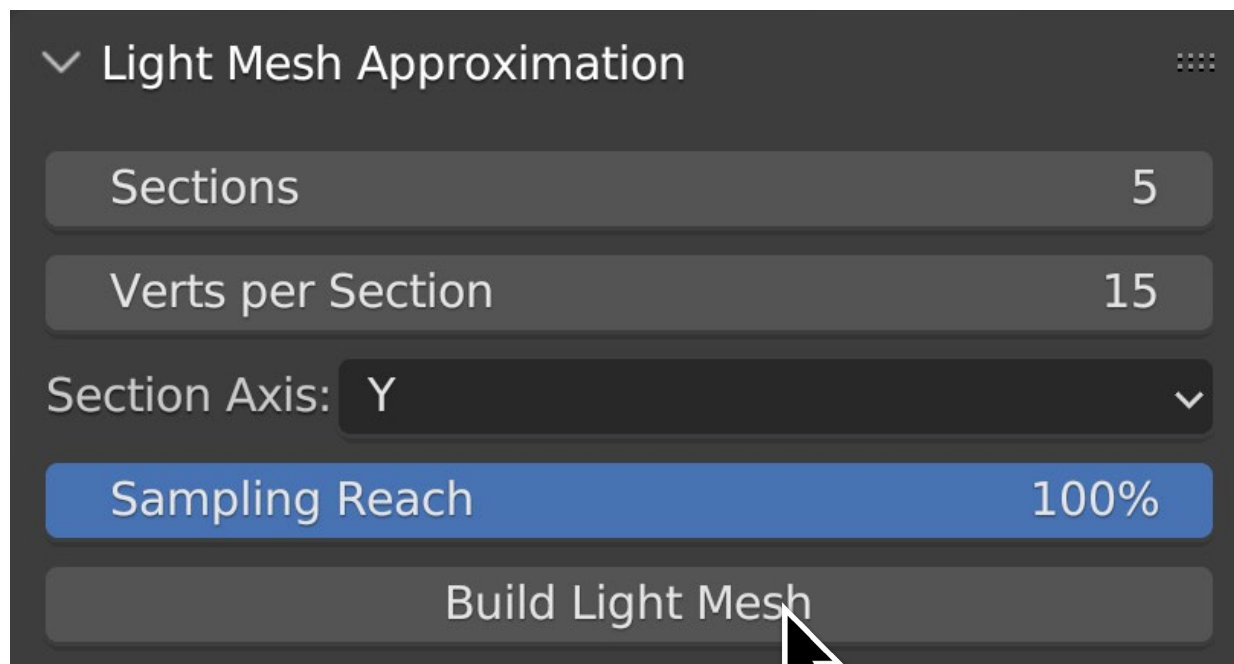
The New Workflow



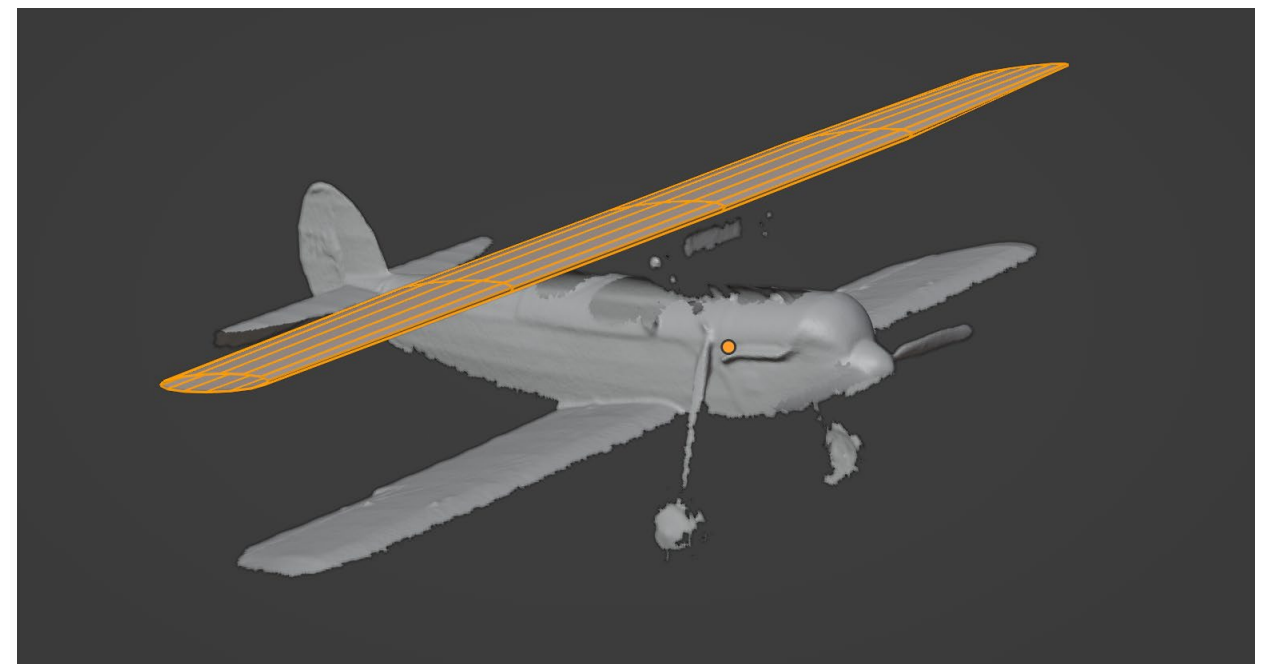
1. Import the heavy 3D scan into Blender



2. Select a chunk of the scan to re-model

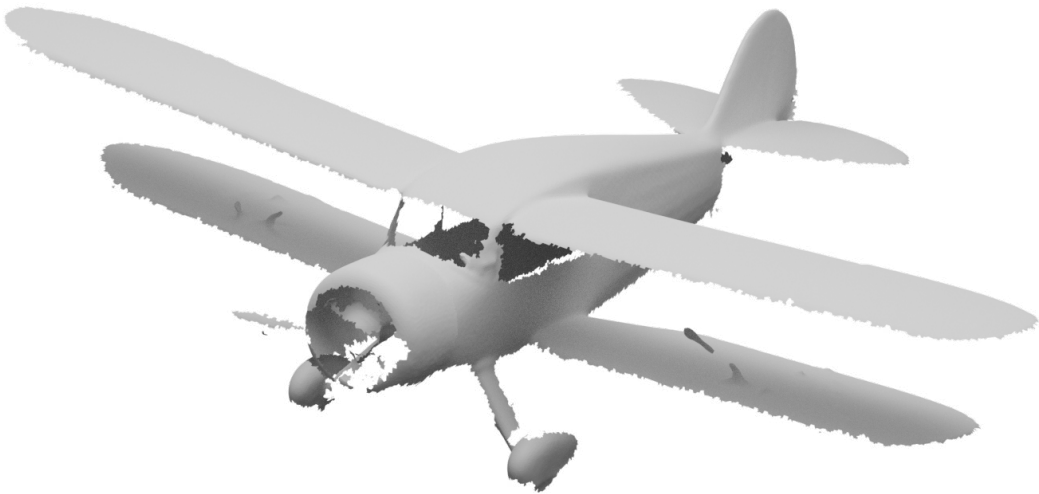


3. Define the output resolution and parameters



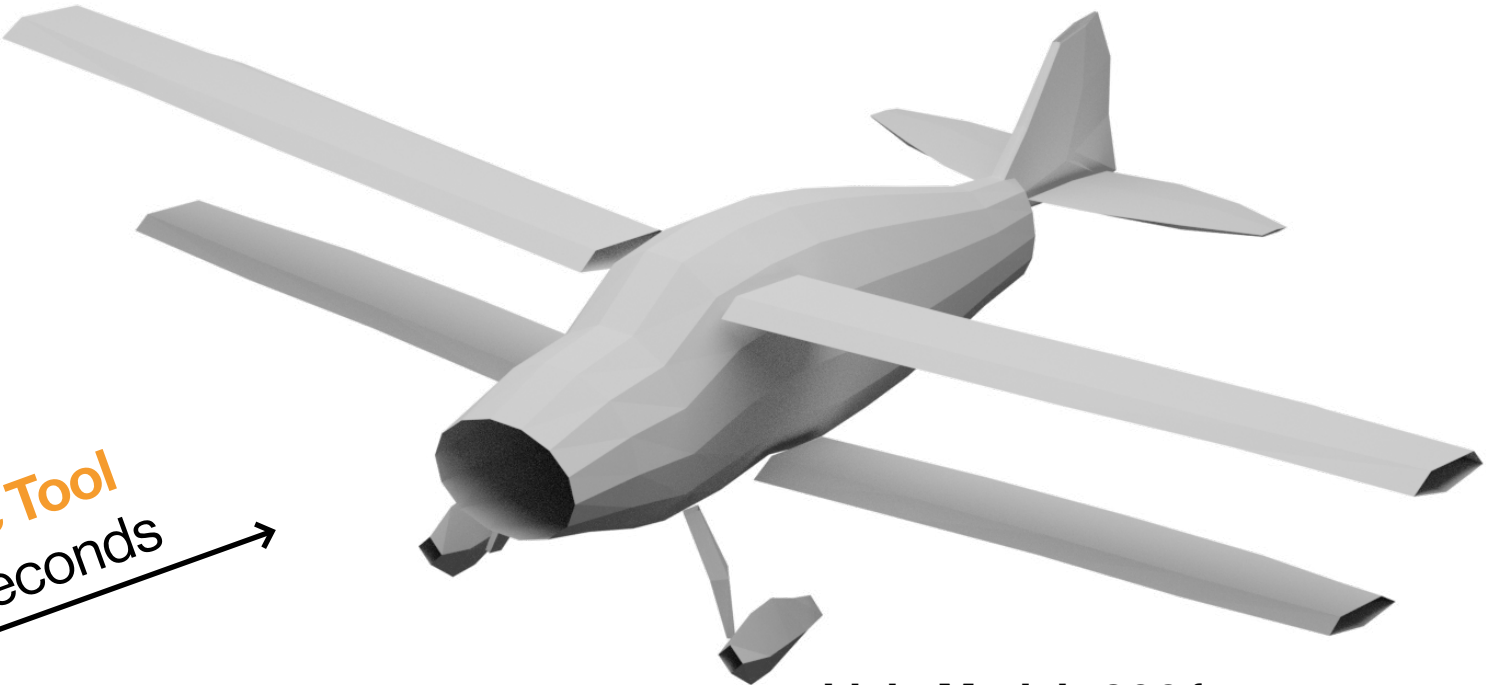
4. Voila! Repeat for all major chunks of object

By the Numbers



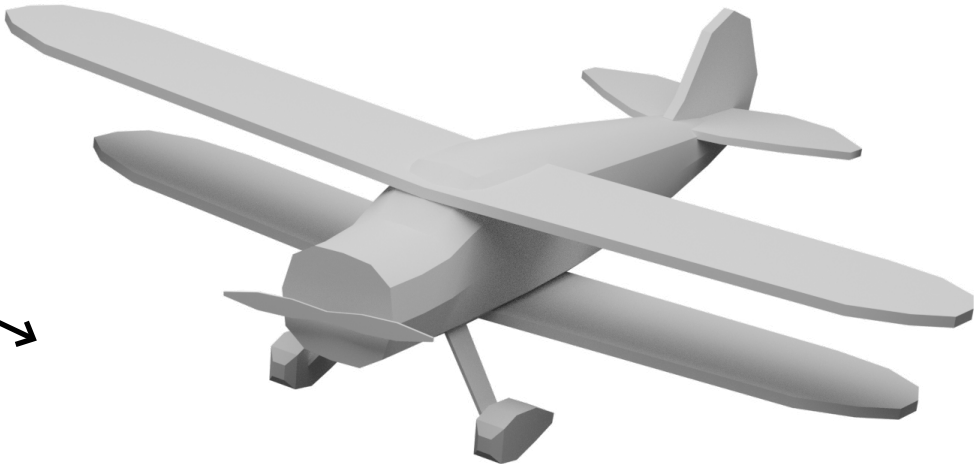
Raw Scan (Waco 10 Scale Model)
389,749 faces

With Automatic Tool
4 minutes 52 seconds



Light Model - 392 faces

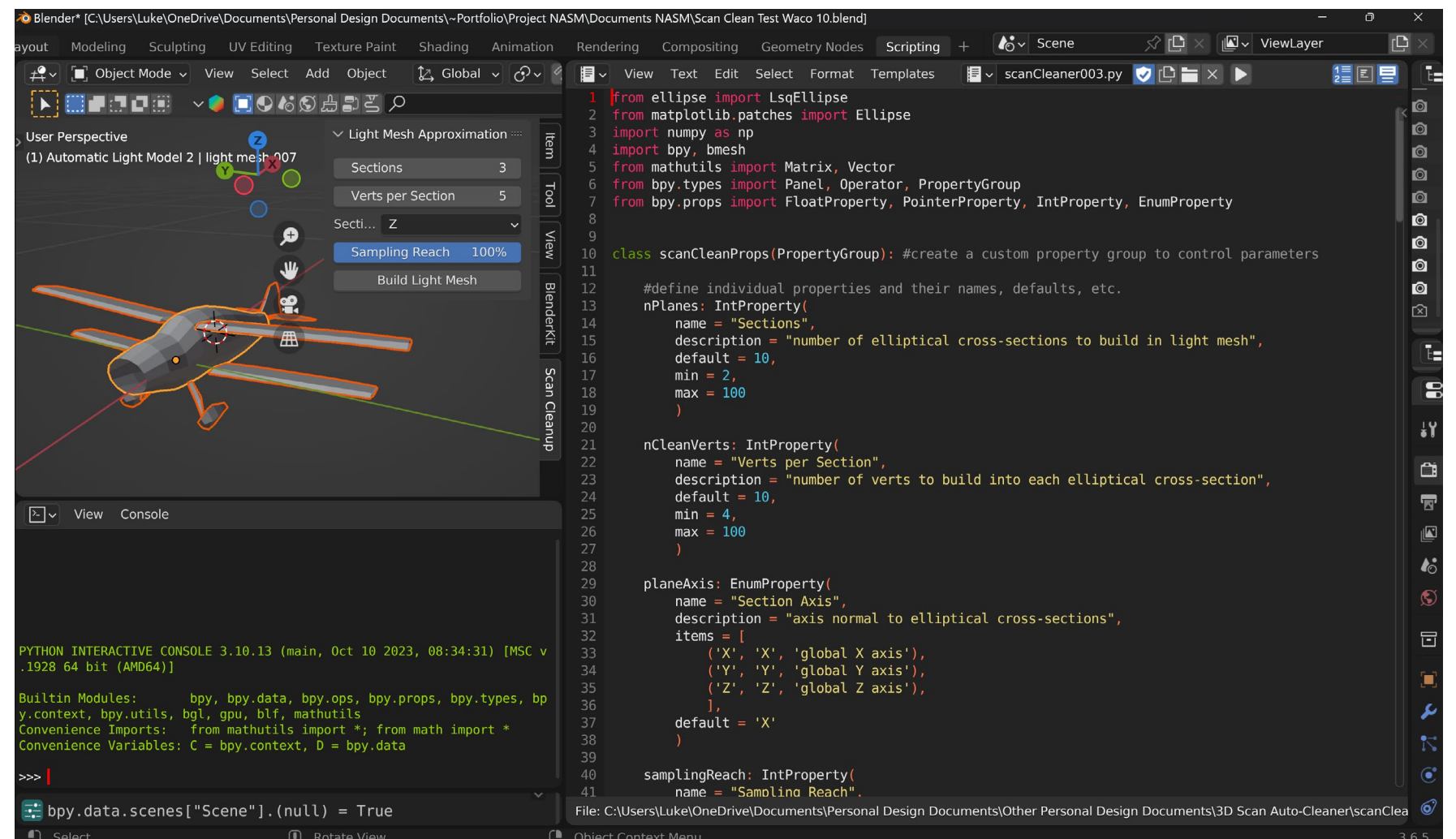
Manual Workflow
13 minutes 44 seconds



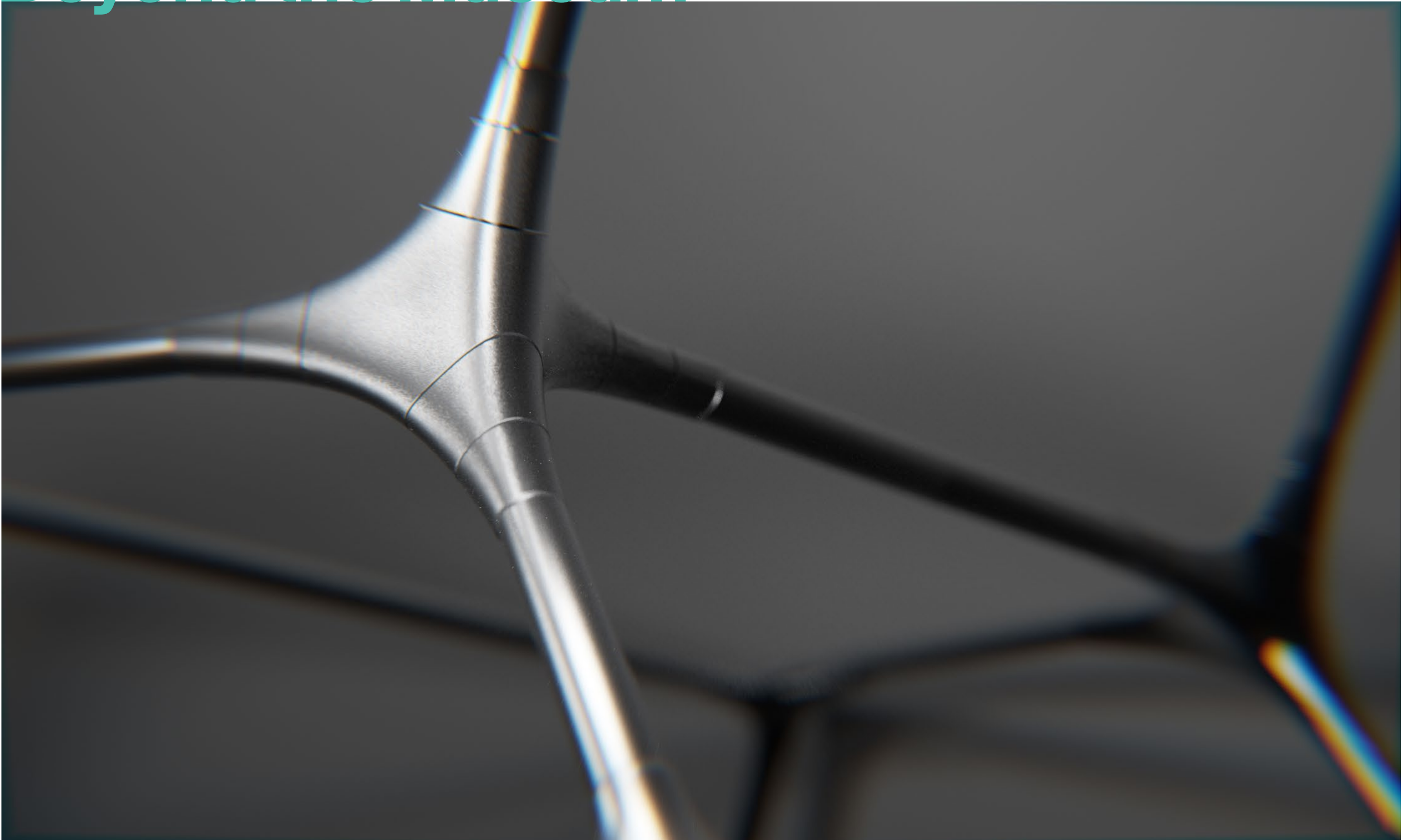
Light Model - 308 faces

While the tool must still be perfected, it promises to produce **light, dimensionally-accurate models faster and with less training** than the current manual workflow.

Imagine what other problems could be solved with tailor-made tools like this!



Beyond the Museum



My skills in other projects

Project 5: **Node Chair**

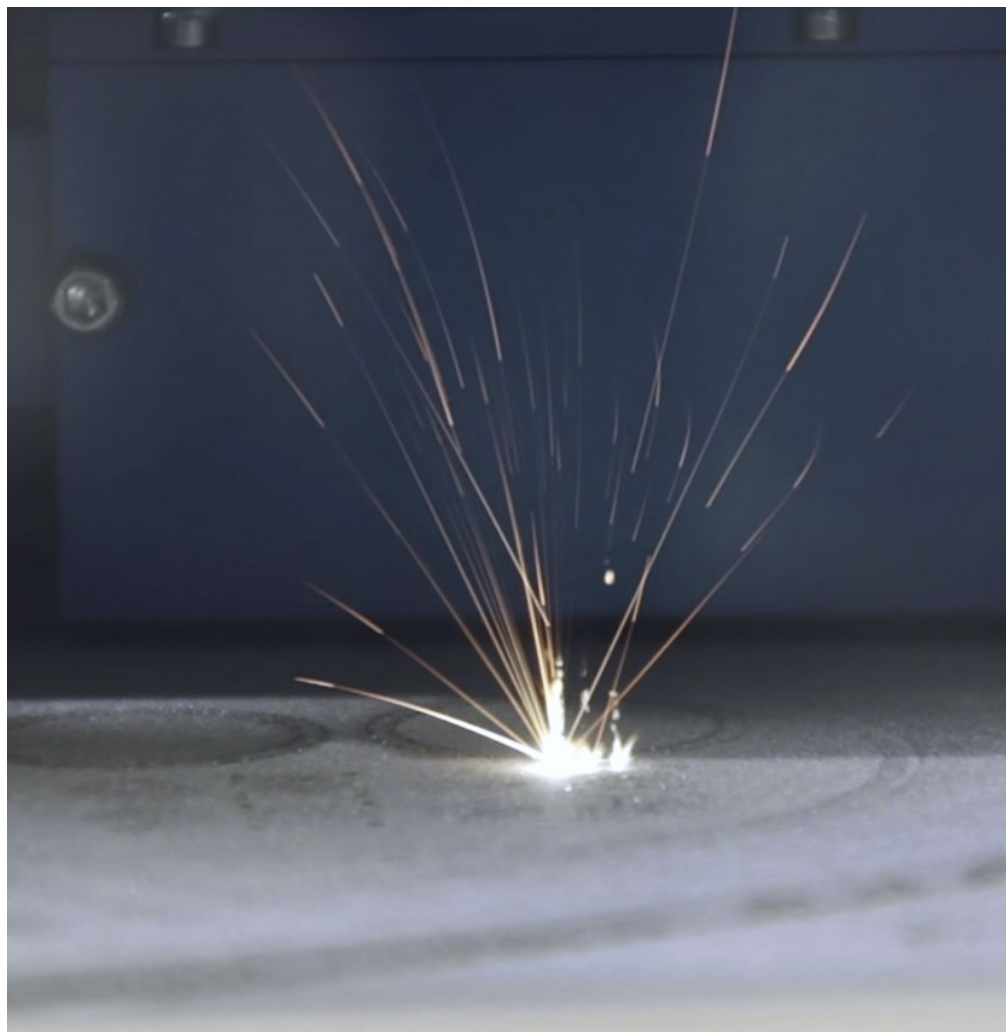
A new furniture paradigm
for tomorrow's manufacturing technologies.



How might we play
to the **unique strengths** of
metal additive manufacturing
through thoughtful design?

↖ My answer

A Solution in need of a Problem



↑ Selective Laser Sintering (SLS) is a form of **metal additive manufacturing (3D printing)**, an up-and-coming category of processes.

Metal 3D printing

is exciting because of its **advantages:**

- Freedom of part geometry
- Easy design changes
- Minimal material waste

but adoption is slow due to its **disadvantages:**

- High cost per part
- Difficult to scale
- Limited tolerances & surface quality

Precedents in Metal 3D Printed Furniture



Immature Technology

MIT researchers have devised a much faster metal additive process, which may make mass production more feasible. However, this technology is in its infancy and can't yet product pieces of suitable quality. **I sought a solution for current additive manufacturing technologies.**

Monolithic Sculptures

Most existing metal additively manufactured chairs are one-part pieces designed for short-run or even one-off production. While some are beautiful, this approach is **not scalable for mass production**. Additive manufacturing is time-consuming, so a completely 3D printed chair is very expensive to make in large numbers.



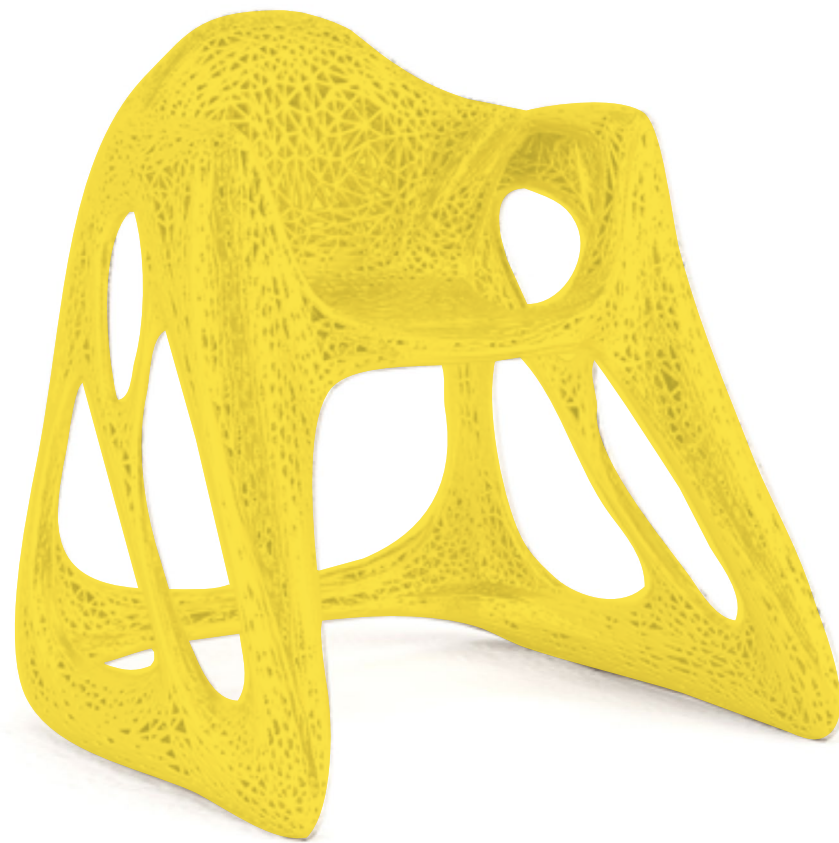
Success in Another Industry

German company Gravity Components mass-produces 3D printed titanium bike components, proving that **an additive product can be successful** if it's the right kind of exotic.



A Different Approach

Precedents



By **using additive manufacturing strategically** where its strengths are most valuable - and using other, more efficient processes elsewhere - Nodule stays cheaper and quicker to manufacture while still leveraging the additive advantage.

Node Chair



- Additively Manufactured
- Fabric
- Cut-to-Length Tubing



It's all about
the nodes.

The metal 3D printed Nodes
lend the chair its
exotic looks,
exceptional assembly experience, and
infinite customizability.

A Proven Concept



This nylon prototype demonstrates the feasibility of the proposed **twist-connect mechanism**.

A knob, cam, and flexible jaws are all **printed directly into the part** in a single process.

Make it **Yours** to Make it **Last**



Since the design of 3D printed parts can change freely,
each Node chair can be customized
to fit the user's space or body. When a
chair is uniquely yours, you're
more likely to repair it when it breaks.
And since Node chairs are
easy to disassemble and repair,
they stay out of the landfill.

Project 6:
Cosmic Reflections

A costume exploring space travel
and its impacts on human minds and culture

The beauty of the cosmos is reflected in us

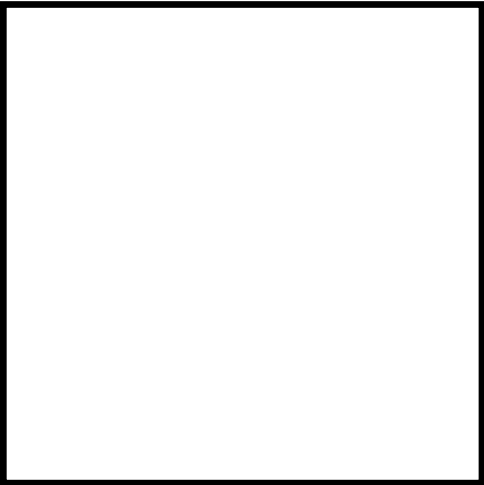
[Watch the Video](#) ►



Model [Erin Secosky](#)

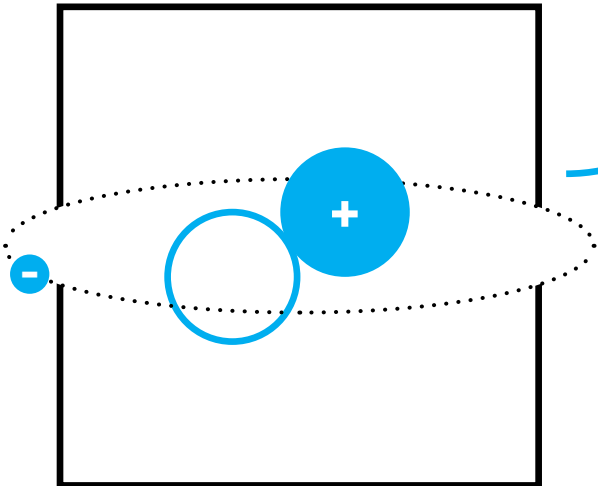
Timeline of the Universe

Initial Expansion



Not much going on.

First Hydrogen Atoms



Homo Sapiens Appear



Made with the
same Hydrogen atoms!

People go to Space



(And are amazed)

Space in Fashion

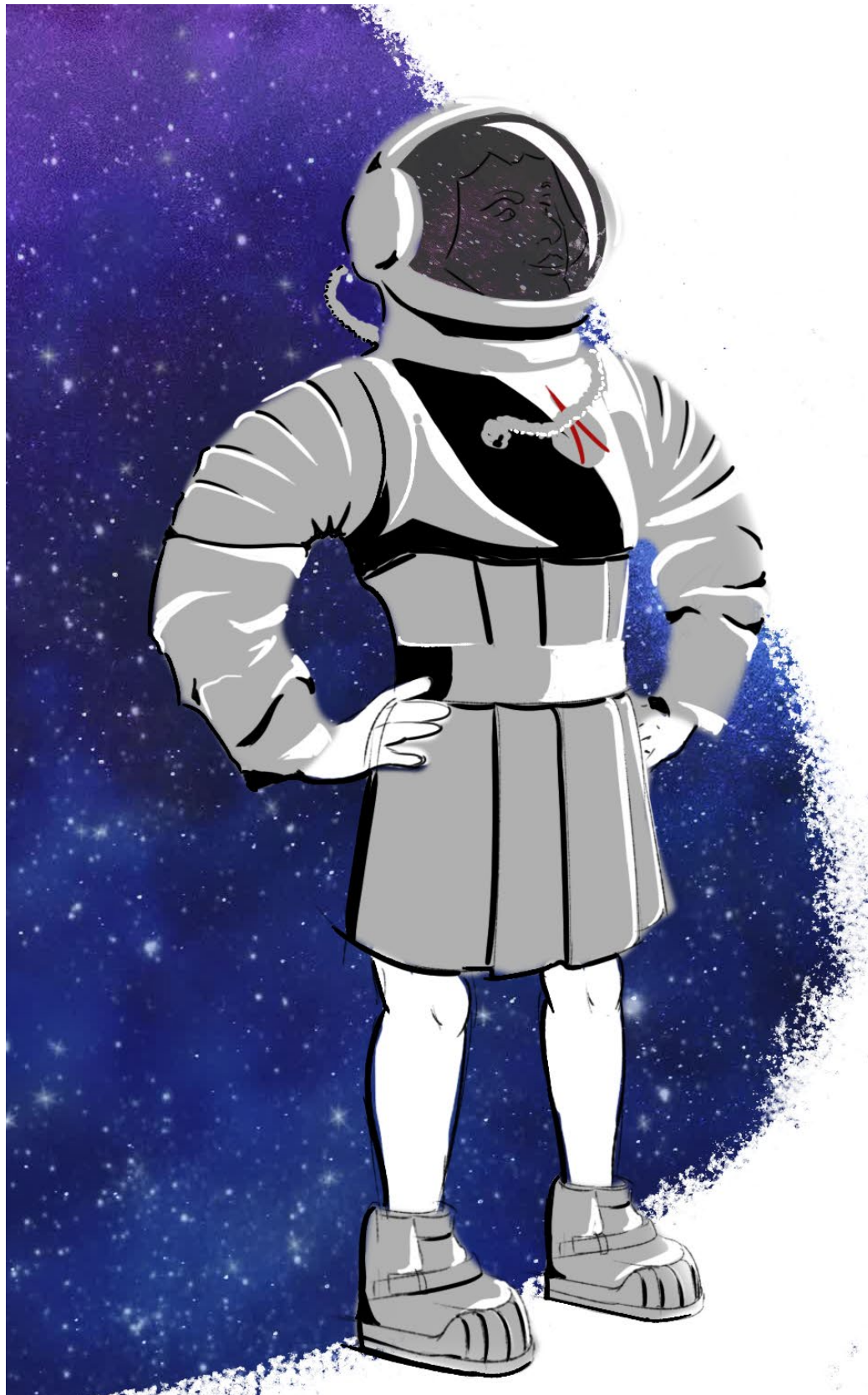


Space invades 60's culture as people are captivated by the cosmos of which they're a part. **Lots of shiny stuff and miniskirts.**



Time since Big Bang (Not Remotely to Scale)

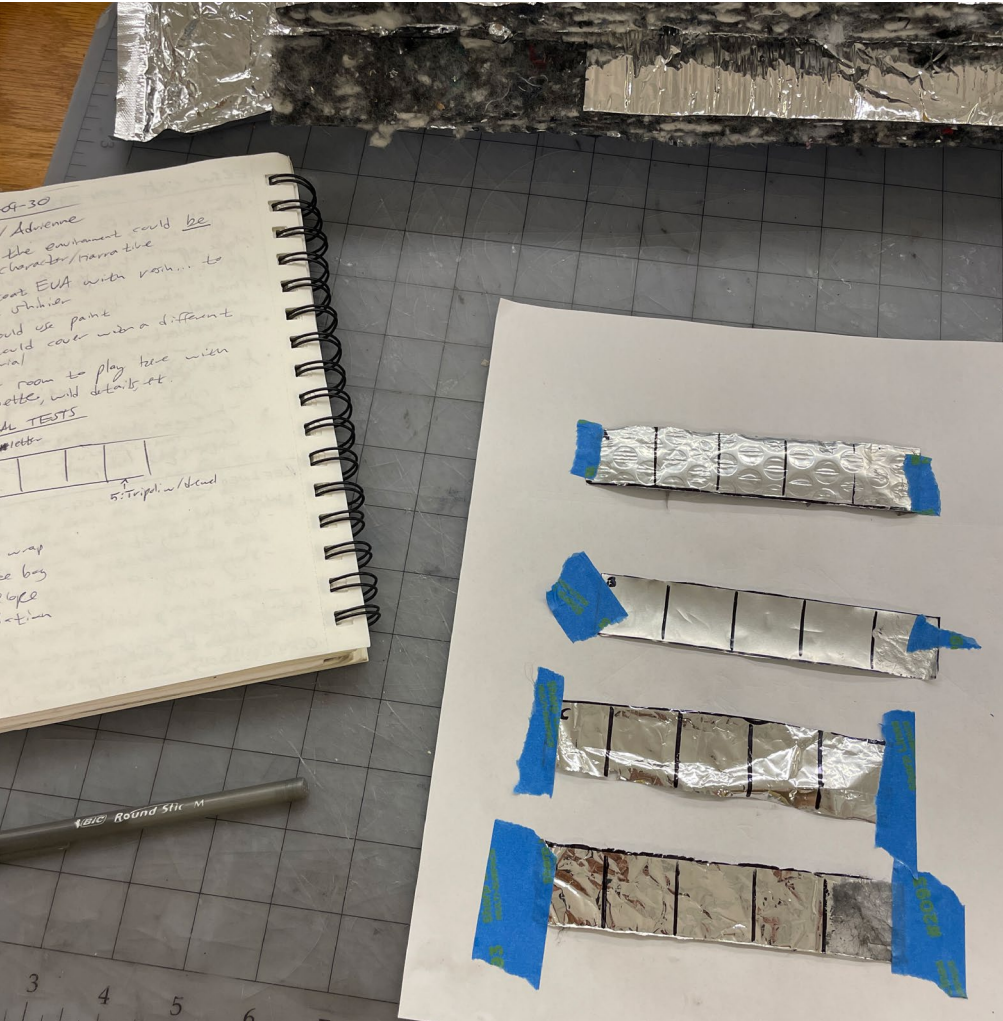
Images: The Austrailian Museum, NASA, New York Times



Images: CNN, NBC News, W Magazine

Thematic inspiration came from the **reflectivity of early spacsuits**, and the **reflection of space travel in space-age fashion**. Details were also drawn from garments of the era.

In Search of Shiny



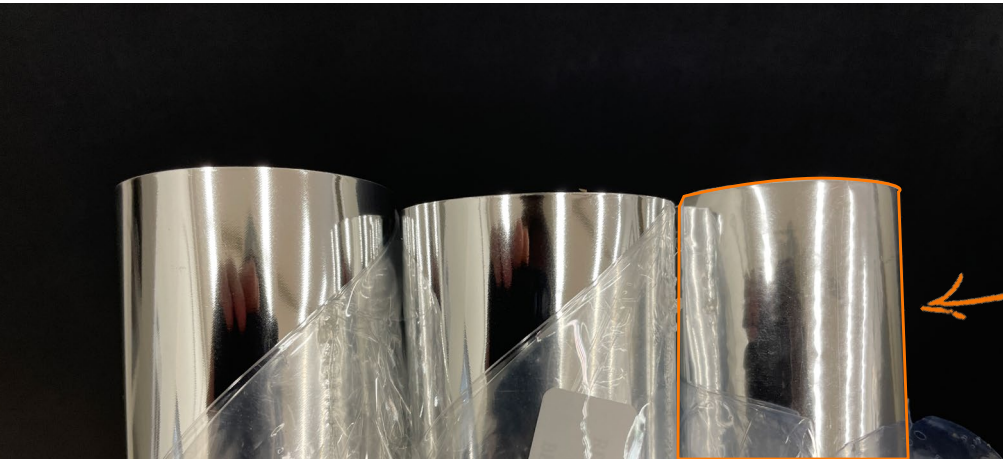
Different packaging materials were **evaluated for shininess** with different finishes.



3D Printed Helmet Models



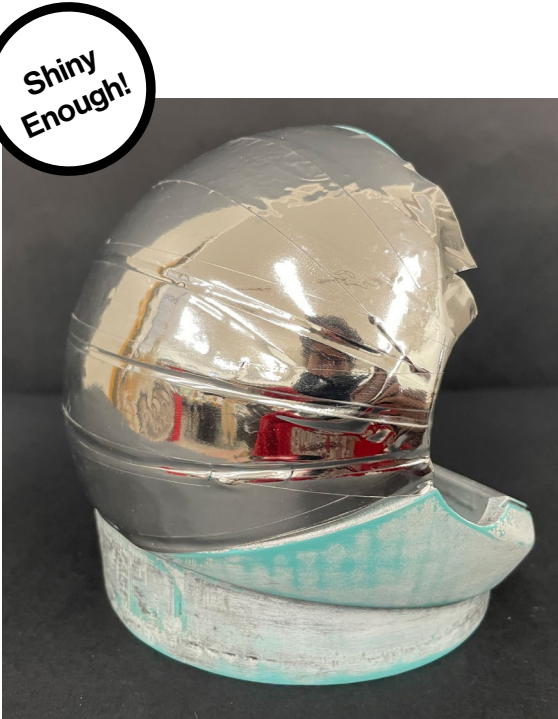
Aluminium Foil Tape



Chrome vinyl is shiny enough. But not *all* chrome vinyl. **Inferior suppliers** were not used.

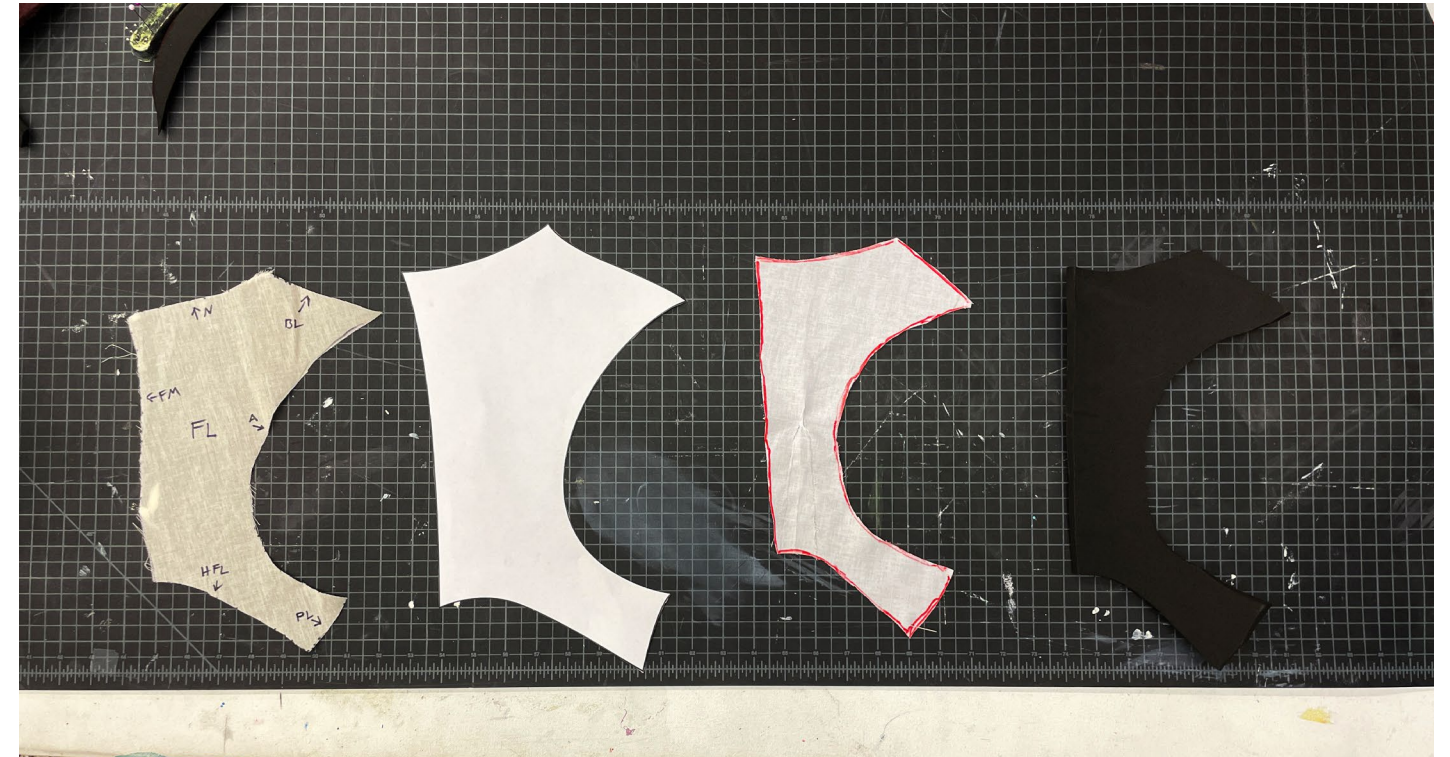


Chrome Paint



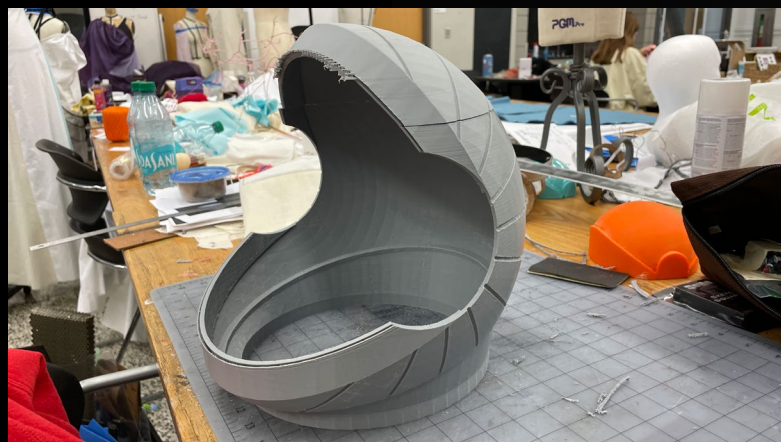
Chrome Vinyl

Design through **Iterative Making**





Thermoformed
Visor



3D Printed
Helmet Shell



Machined
Aluminium
Details





EVA Foam
Upper Torso



Sheet Metal
Interface Rings



Sewn
Arms & Skirt



Designed for **Wearability**



Detachable hose ends

Velcro-on arms with flexible interfaces

Integrated lightswitch

Zipper closure

All to make the suit easier to don, doff and wear.



Assistant Photographer Max Decker

Images from the James Webb Space Telescope remind us how beautiful the universe is. Let them also remind us how remarkable it is that we were born from it.

We are part of **The Most Interesting Thing that has Ever Happened.**

“If you think galaxies are amazing, just wait until you find out about you... [it] does not seem very common for the universe to find paths towards waking up. And to be a part of that, it feels very big.”

-Hank Green



Last Summer



A look back: 10 weeks at NASM

Project 2:

Gallery 211 **Rendering**

These visuals of the under-renovation art gallery facilitated promotion of the project by leadership. The gallery had already been designed; I polished material and lighting effects and rendered it.





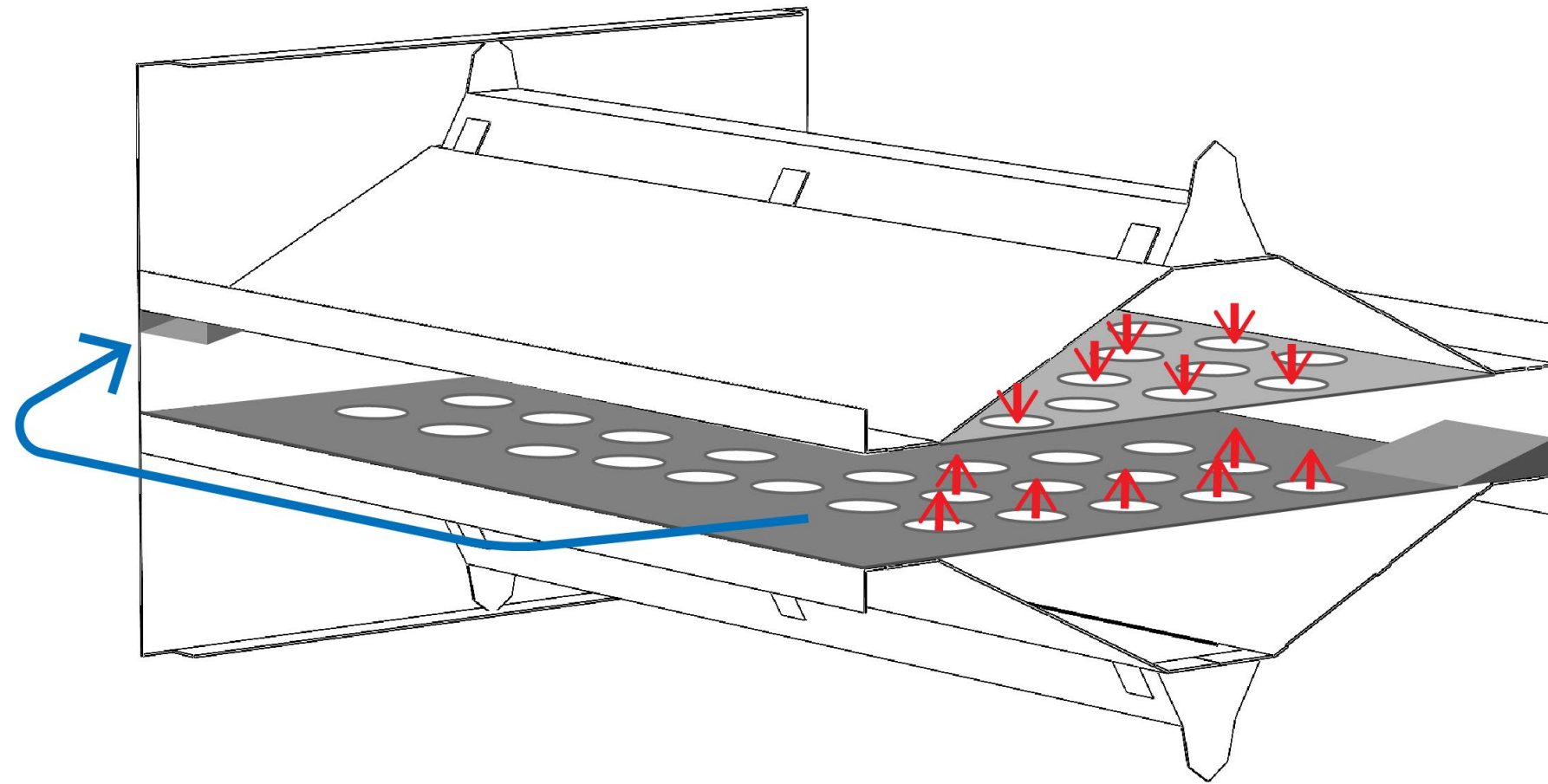




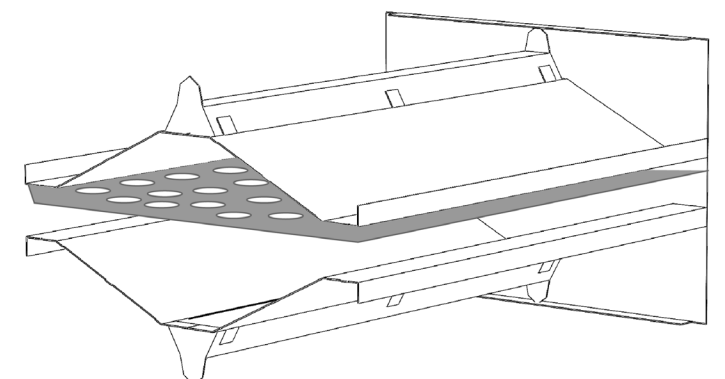
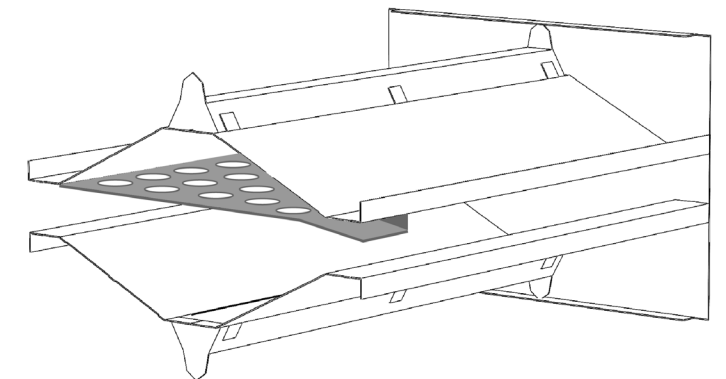
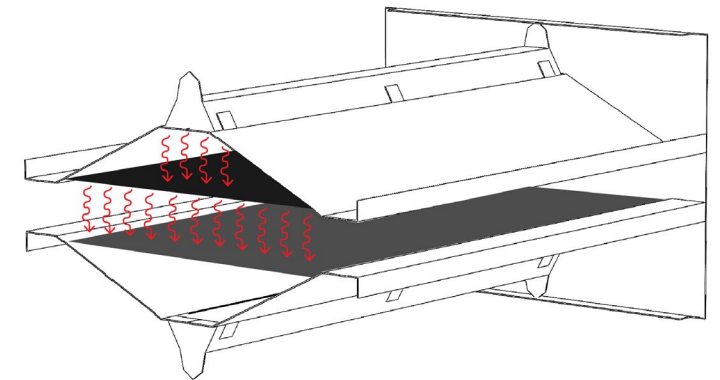
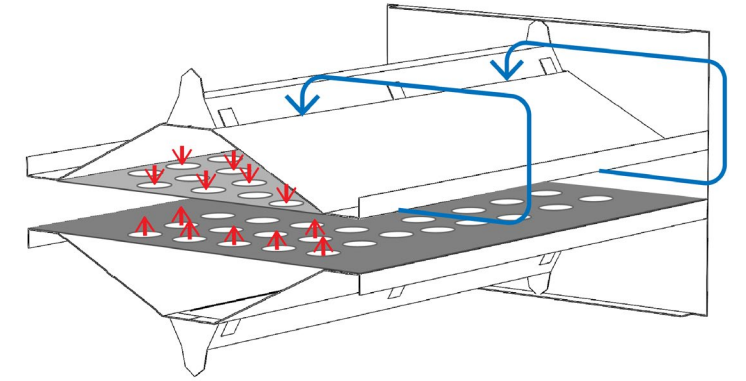
Project 3: Thermoforming Oven **Fabrication**

Exhibits Production needed a way to reliably shape large pieces of Corian and other materials into curves. Tenzin Phuntsok and I designed and fabricated a large thermoforming oven to evenly heat material for forming.

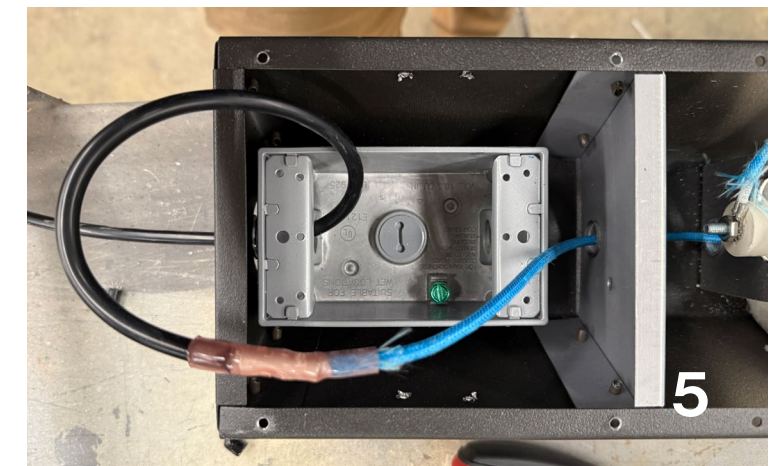
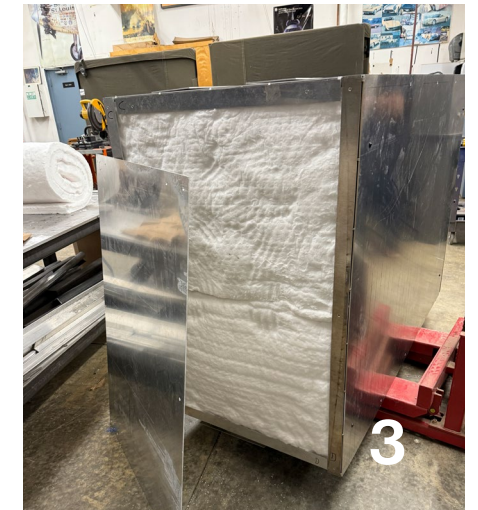
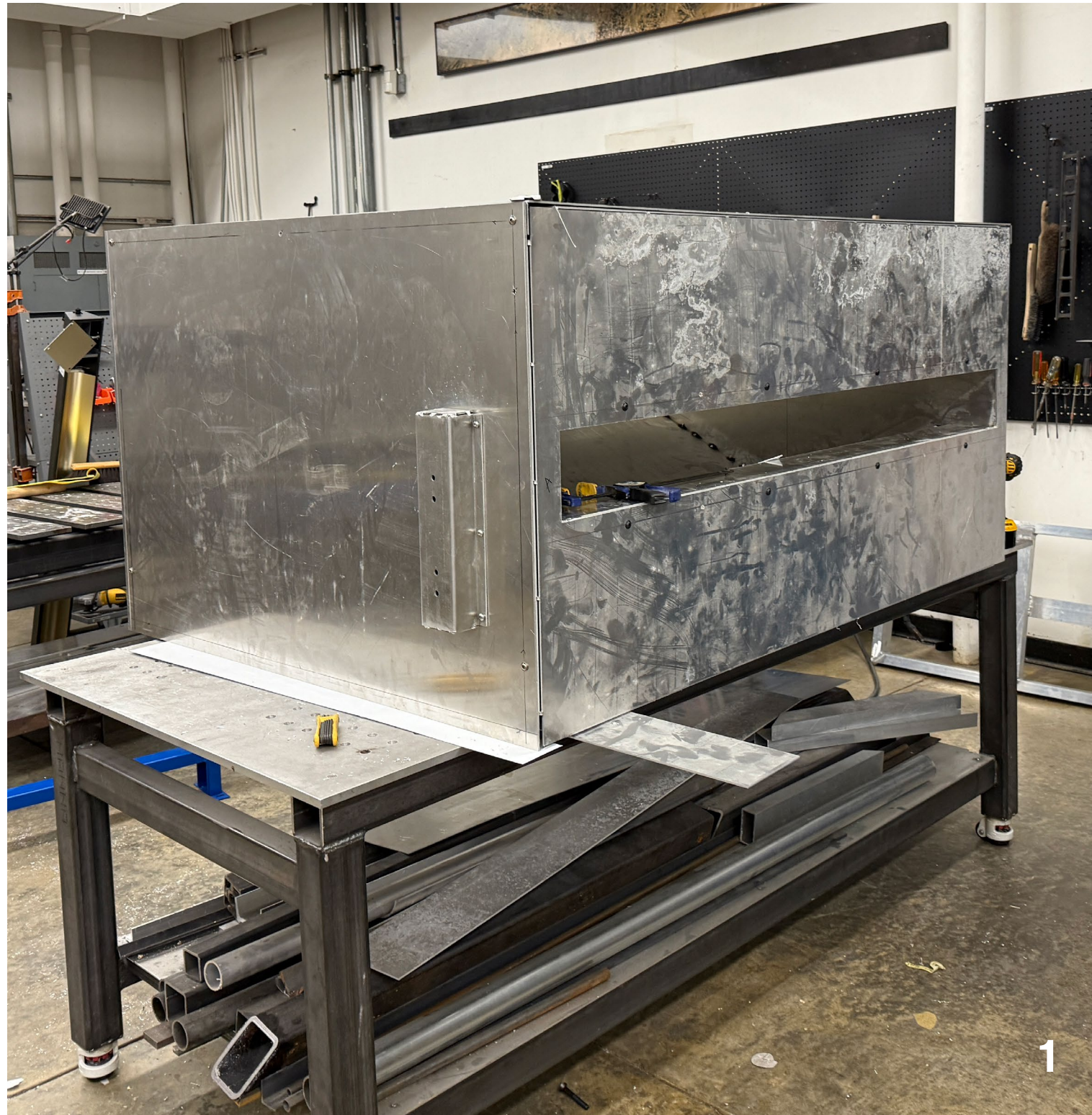
Thermal Considerations



To promote even heating of the material, we explored a variety of designs which prioritized **convective and radiant heat transfer**. I created these visualizations to help consider the implications of these for the structure of the oven.

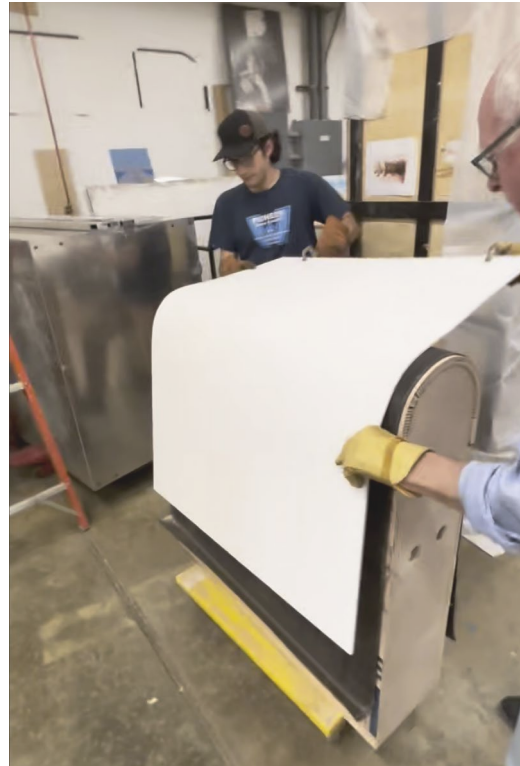


Fabricating the Oven



- 1** The oven nearly ready for testing
- 2** Bending waterjet-cut aluminum panels, to be welded or drilled, tapped and bolted together.
- 3** Insulating the oven for efficiency and safety
- 4** Soldering connections for the heaters and thermostats.
- 5** Finished, safe and tidy electrical connections

Testing the Oven



While the process must still be perfected, initial testing demonstrated the oven's ability to turn a **rigid sheet of Corian** into a **pliable, moldable surface**.

Project 4: TipTap Donation **Concept Development**

Tap-to-pay technology was implimented as a smart, easy way for visitors to donate to the museum. But the existing implimentation was underperforming. I teamed up with BOT interns to find a solution.

TipTap: The Problem



FY 2024 Donation Revenue

Store Register



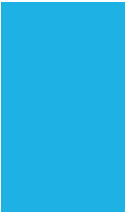
Entry Ticket Donations



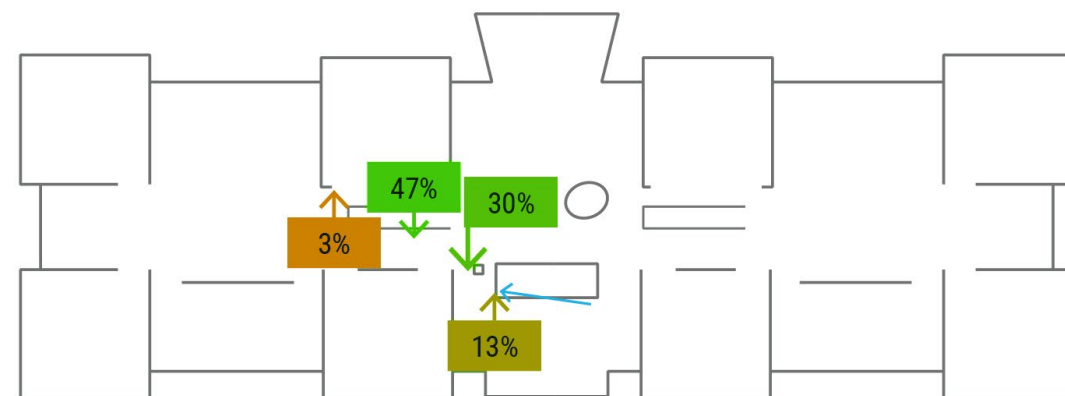
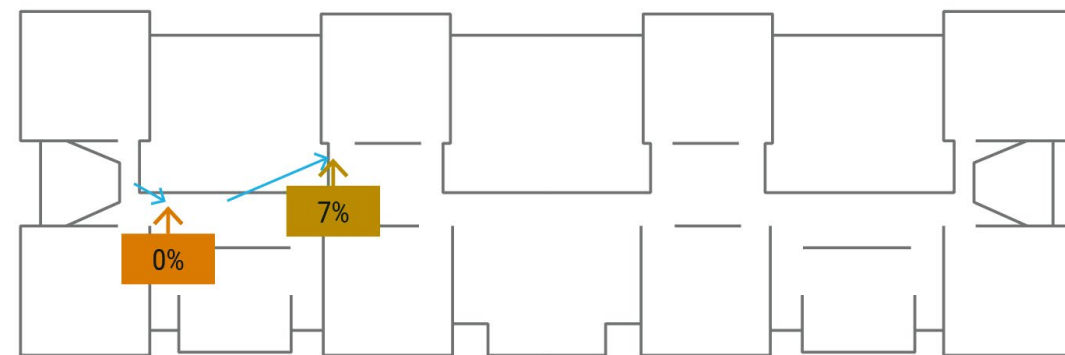
Cashboxes



TipTap



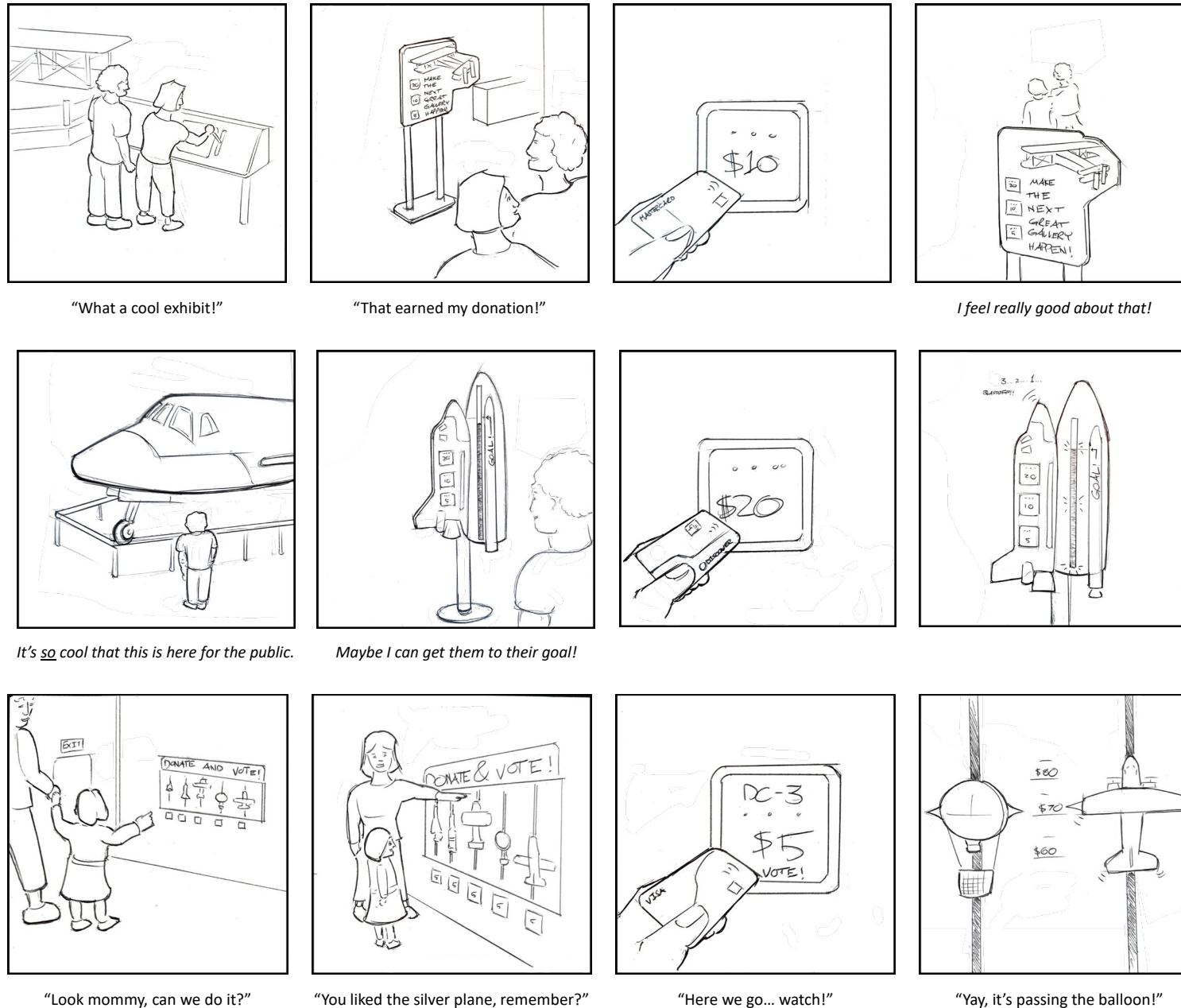
TipTap: The Process



↑ I organized a **design thinking workshop** to generate solution ideas. It's a building full of smart people - why not use them?

← **Location research** revealed the power of placement in likelihood that a visitor would donate.

TipTap: The Process

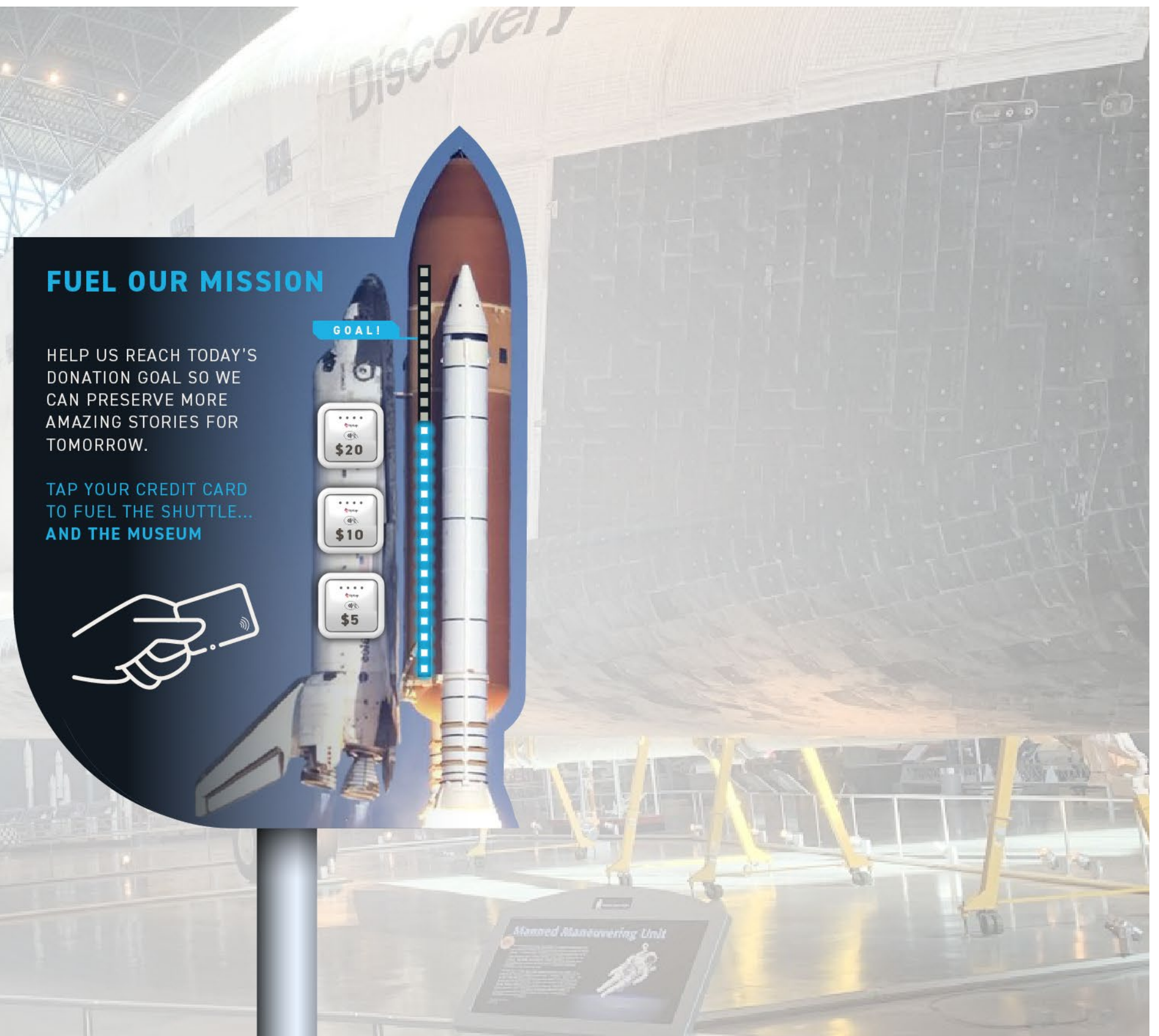


Visitor experience was at the core of the entire design process, from **storyboards** to **interviews** and more.

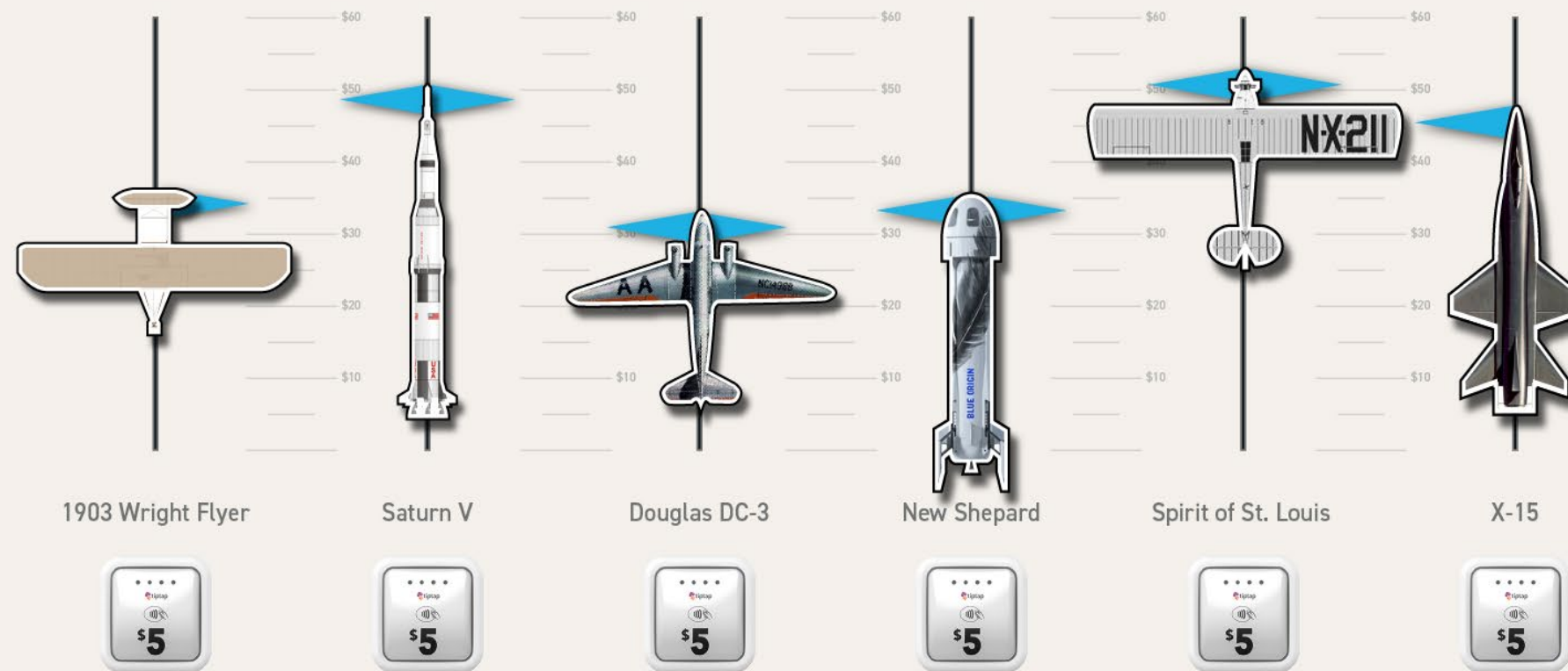
Concept 1: this display motivates visitors to give by emphasizing the power of a donation to build more great exhibits, including popular interactive elements. Its relation to a popular artifact draws attention.



Concept 2: in addition to showing visitors the impact of their gift, this interactive display offers a small “value exchange” - a satisfying result when a visitor donates. People are also more likely to donate when they can see that others have.



DONATE AND VOTE FOR YOUR FAVORITE ARTIFACT!



TAP A CARD TO DONATE AND VOTE



Concept 3: the “value exchange” effect is enhanced by offering the visitor a voice, and an opportunity to leave their mark on the museum.

Let's make cool stuff together.

📷 @RamsdellRandD

✉ luke2412d@gmail.com

📞 +1 (828) 773-3564

